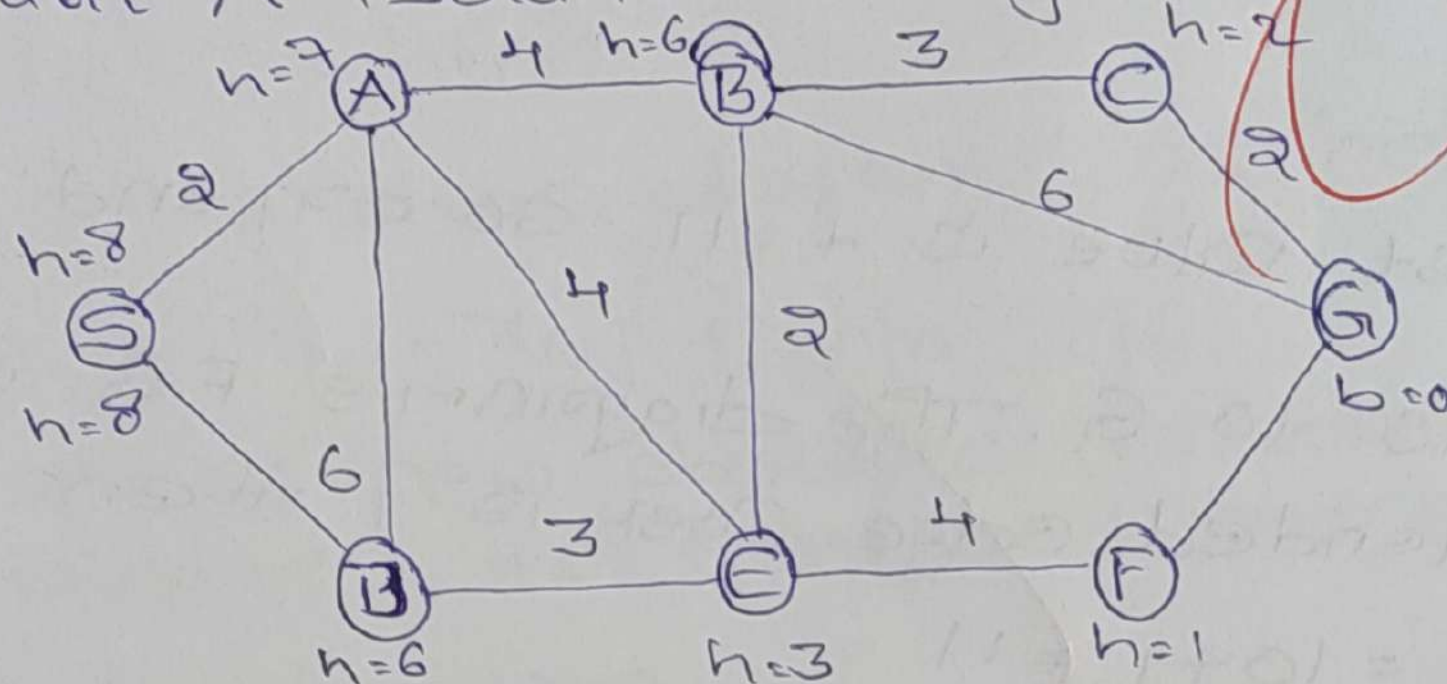


27/02/26

1) Smart Ambulance Navigation:



We use the A* evaluation function:

$$f(n) = g(n) + h(n)$$

where,

 $g(n)$ = actual cost from S to node n. $h(n)$ = heuristic estimate from n to G $f(n)$ = total estimated cost

1. Initial node

for S:

$$g(s) = 0 ; h(s) = 8$$

$$f(s) = 0 + 8 = 8$$

Expand S:

$$S \rightarrow A = 2$$

for A,

$$g(A) = 2 ; h(A) = 7$$

$$f(A) = 2 + 7 = 9$$

2. Expand A:

Node	Edge cost	$g(n)$	$h(n)$	$f(n) = g + h$
B	4	6	6	12
D	6	8	6	14
E	4	6	3	9

The smallest value is $E = 9$, so expand E from E

The smallest value is $f = 11$, so expand F.

x. F connects to G. The diagram's F-G edge label is the intended edge cost is 1, then

$$g(G) = 10 + 1 = 11$$

$$h(G) = 0$$

$$f(G) = 11 + 0 = 11$$

Selected Route:

$$S \rightarrow A \rightarrow E \rightarrow F \rightarrow G$$

Cost:

$$2 + 4 + 4 + 1 = 11$$

sequence of node explored

$$S \rightarrow A \rightarrow E \rightarrow F \rightarrow G$$

Hence,

Algorithm: A* search

$$\text{Formula: } f(n) = g(n) + h(n)$$

selected route: $S \rightarrow A \rightarrow E \rightarrow F \rightarrow G$.

Total Cost: 11

Optimal: yes

Autonomous warehouse Robot.

Given,

* Start (S) : Col 1, Row 5

* Goal (G) : Col 7, Row 1

Search Strategy:

The robot uses online A^* :

$$f(n) = g(n) + h(n)$$

* Use Manhattan distance as heuristic.

$$h(n) = |x_n - x_G| + |y_n - y_G|$$

whenever a new obstacle is detected, the robot updates its map and runs A^* again.

1. Initial decision.

The robot starts at:

$$S = (1, 5)$$

Goal:

$$G = (7, 1)$$

A possible initial route

$$S \rightarrow (2, 5) \rightarrow (3, 5) \rightarrow (4, 5) \rightarrow (5, 5) \rightarrow (6, 5) \rightarrow (7, 1)$$

2. Obstacle discovery.

When the robot reaches (5, 5), it observes the neighbouring cell: (5, 4)

It discovers that

$$(5, 4) = \#$$

Then A^* executed again.

New route:

$$(5, 5) \rightarrow (6, 5) \rightarrow (7, 5) \rightarrow (7, 4) \rightarrow (7, 3)$$

3. Another obstacle is discovered.

When the robot reaches (7, 4), it detects

$$(7, 3) = \#$$

There, the robot can't move directly upwards to the goal.

The new route becomes
 $(7,4) \rightarrow (6,4) \rightarrow (6,3) \rightarrow (6,2)$

At $(6,3)$ the robot discovers
so this route is also blocked

A^* now finds:

$(6,3) \rightarrow (5,3) \rightarrow (5,2) \rightarrow (5,1)$

Then,

$(5,1) \rightarrow (6,1) \rightarrow (7,1)$

and reach the goal.

$\therefore$ one valid online search execution is

$S \rightarrow (2,5) \rightarrow (3,5) \rightarrow (4,5) \rightarrow (5,5) \rightarrow (6,5) \rightarrow (7,1)$

The robot reaches the target after planning
whenever an obstacle is discovered.

3.) University Examination Scheduling

a). CSP Formulation

Variables:

$E_1, E_2, E_3, E_4, E_5, E_6$

Domain:

$\{T_1, T_2, T_3, T_4, T_5\}$

Constraints:

* $E_1 \neq E_2$

* $E_1 \neq E_3$

* $E_2 \neq E_4$

* $E_3 \neq E_5$

* $E_4 \neq E_6$

* $E_2 \neq E_5$

Each examination must also be assigned a hall
whose capacity is sufficient.

b. MRV + Forward checking
 using MRV, choose the variable with the smallest remaining domain. For ties, choose the exam having more conflicts.

A valid assignment is

Step	Exam	Assigned slot
1.	E_2	T_1
2	E_1	T_2
3	E_4	T_2
4	E_3	T_3
5.	E_5	T_4
6	E_6	T_1

c. Domains After each Assignment

After Assign

$$E_2 = T_1$$

$$E_1 = T_2$$

$$E_4 = T_2$$

$$E_3 = T_3$$

$$E_5 = T_4$$

$$E_6 = T_1$$

Final table:

Time

Examination

1
1
1
2
2
3

E_2

E_6

E_1

E_4

E_3

Remaining domains

$$E_1: \{T_2, T_3, T_4\}; E_4: \{T_2, T_3, T_4\}$$

$$E_5: \{T_2, T_3, T_4\}$$

$$E_3: \{T_1, T_3, T_4\}; E_4: \{T_2, T_3, T_4\}$$

$$E_5: \{T_2, T_3, T_4\}$$

$$E_3: \{T_1, T_3, T_4\}; E_5: \{T_2, T_3, T_4\}$$

$$E_6: \{T_1, T_3, T_4\}$$

$$E_5: \{T_2, T_4\}; E_6: \{T_1, T_3, T_4\}$$

$$E_6: \{T_1, T_3, T_4\}$$

Complete

Hall

Students

Hall B

80

Hall C

50

Hall A

100

Hall B

60

Hall A

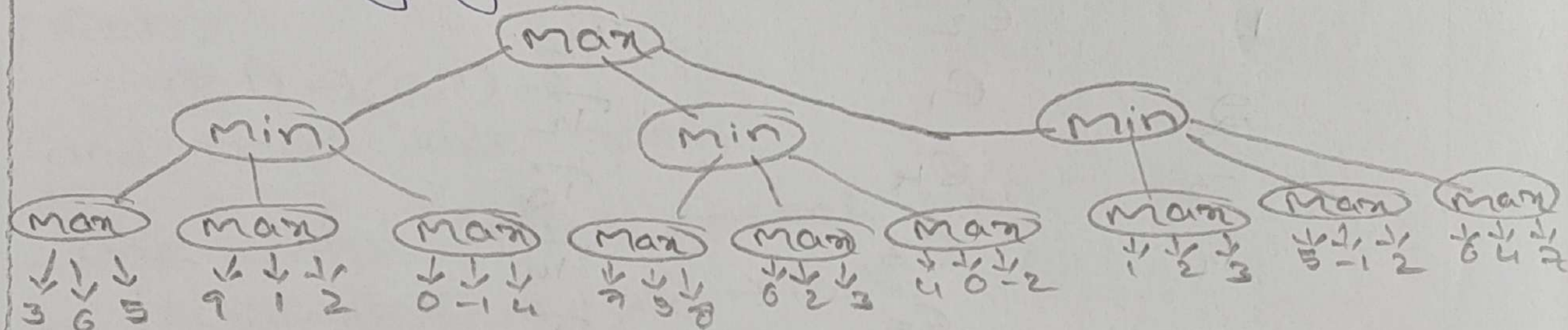
70

Result:

All student-Conflict and hall-Capacity Constraints are satisfied

MKV + forward checking gives a valid examination timetable without backtracking.

4.) Game-Playing A2:



Minimax.

The root is a Max node, and the next value at each Max node:

$$* \text{Max}(3, 5, 6) = 6$$

$$* \text{Max}(9, 1, 2) = 9$$

$$* \text{Max}(0, -1, 4) = 4$$

Left min:

$$\text{Min}(6, 9, 4) = 4$$

Middle:

$$* \text{Max}(7, 5, 8) = 8$$

$$* \text{Max}(6, 2, 3) = 6$$

$$* \text{Max}(4, 0, -2) = 4$$

$$\text{Min}(8, 6, 4) = 4$$

Right,

$$* \text{Max}(1, 2, 3) = 3$$

$$* \text{Max}(5, -1, 2) = 5$$

$$* \text{Max}(6, 4, 7) = 7$$

$$\text{Min}(3, 5, 7) = 3$$

These for at the root:

$$\text{Max}(4, 4, 3) = 4$$

final Maximax move

A2 chooses the left or middle branch

Both give score of 4.

Alpha-Beta pruning;

searching from left to right

After evaluating the left and middle branches
the root already has: $\alpha = 4$.

In the right min branch the first max node
 $\text{Max}(1, 2, 3) = 3$

Since $3 < 4$

The remaining branches don't need to be
evaluated.

pruned branches

$(-1, 2)$ and $(6, 4, 2)$

Conclusion:

~~Maximum~~ minimum value = 4

Alpha-Beta pruning removes unnecessary
branches and gives the same answer faster

5) Delivery Route Optimization:

Given routes and Costs

<u>Route</u>	<u>Cost</u>
S1: $D \rightarrow A \rightarrow C \rightarrow B \rightarrow E \rightarrow D$	26 km
S2: $D \rightarrow B \rightarrow A \rightarrow C \rightarrow E \rightarrow D$	30 km
S3: $D \rightarrow A \rightarrow B \rightarrow E \rightarrow C \rightarrow D$	24 km
S4: $D \rightarrow E \rightarrow C \rightarrow B \rightarrow A \rightarrow D$	22 km
S5: $D \rightarrow A \rightarrow E \rightarrow B \rightarrow C \rightarrow D$	23 km
S6: $D \rightarrow C \rightarrow B \rightarrow A \rightarrow E \rightarrow D$	29 km.

1. Optimization strategy

use Hill climbing, a local search techniques.

* Start with the initial route.

* Generate neighbouring routes by changing
the order of customers.

* Calculate the total distance of each neighbour

x. Repeat until no neighbouring route gives a better solution.

2. Step-by-step:

The best route among routes is:

SS: $D \rightarrow A \rightarrow E \rightarrow B \rightarrow C \rightarrow D$

Its Costs is

$$D-A + A-E + E-B + B-C + C-D$$

$$= 6 + 6 + 5 + 4 + 8$$

$$= 29 \text{ km}$$

Correction: The image deals ss as 23 km, but the distance matrix actually gives

$$6 + 6 + 5 + 4 + 8 = 29 \text{ km}$$

Shortest route:

$D \rightarrow A \rightarrow B \rightarrow C \rightarrow E \rightarrow D$

$$6 + 3 + 4 + 3 + 7 = 23 \text{ km.}$$

Final route: $D \rightarrow A \rightarrow B \rightarrow C \rightarrow E \rightarrow D$

4. Is a global optimum?

yes, for the given distance matrix, 23 km is the global optimum.

5. Hill climbing can get stuck at a local optimum. To escape it, we can use.

1. Random restart Hill climbing
2. Simulated Annealing
3. Stochastic hill climbing